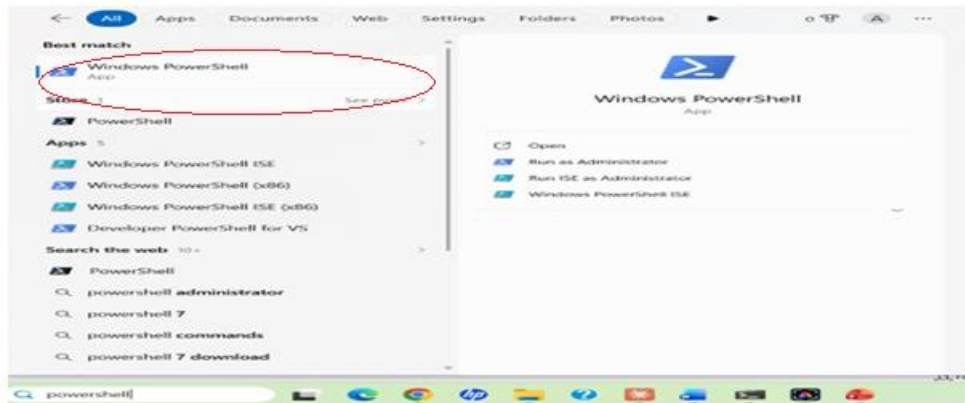
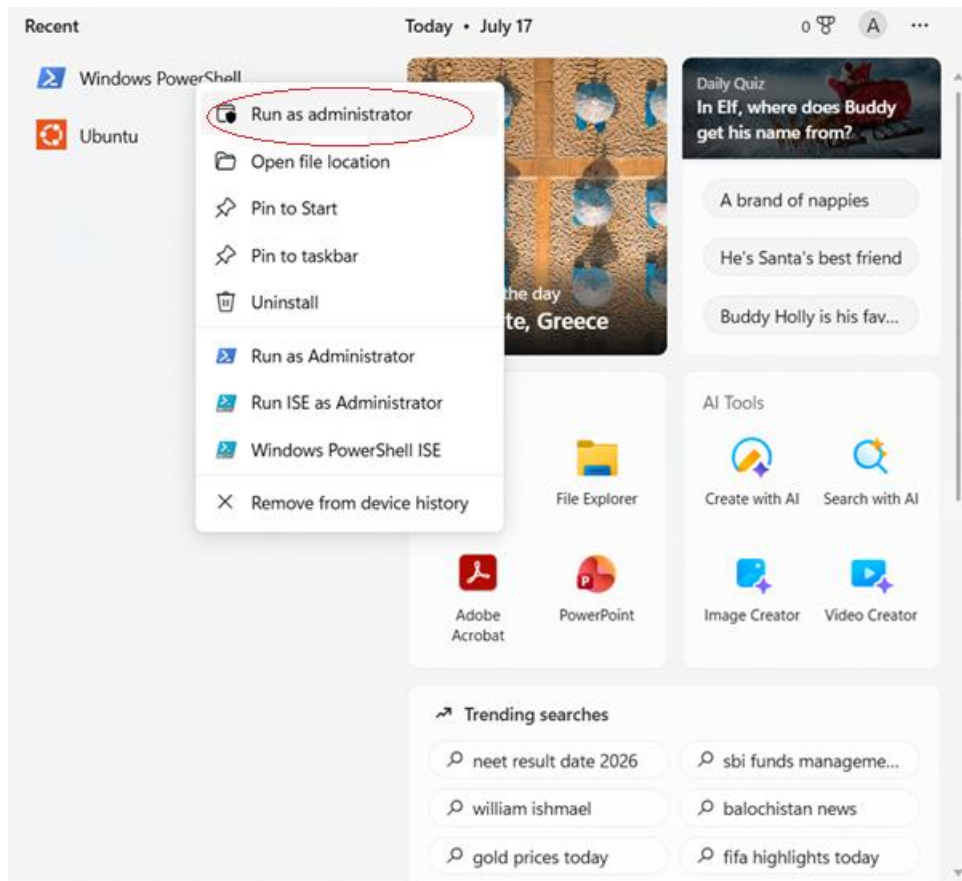


Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.



3. Right-click **Windows PowerShell**.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> wsl --install
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: saisharan134

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
saisharan134@APL-5CD1366KSL:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Hit:12 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Fetched 3159 kB in 3s (1022 kB/s)
52 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC (C Compiler)

```
saisharan134@APL-5CD1366K5L:~$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 52
saisharan134@APL-5CD1366K5L:~$
```

Step 8: Verify the Installation

```
saisharan134@APL-5CD1366K5L:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

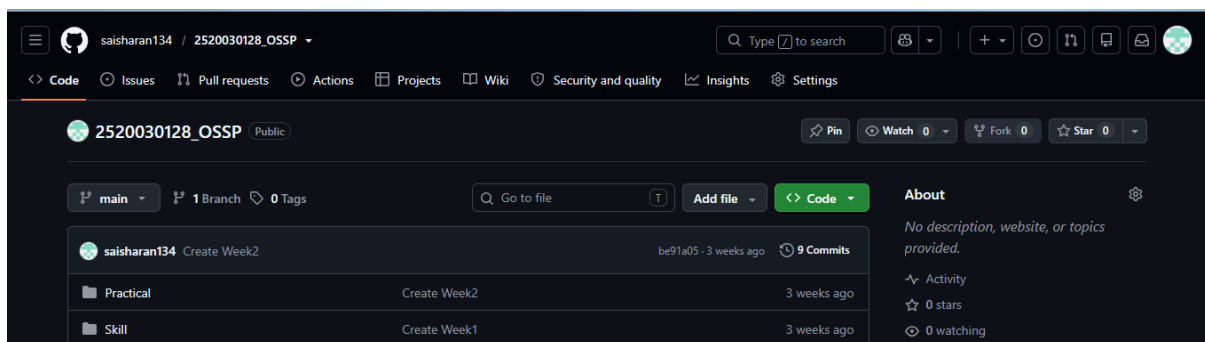
Step 9: Create Your C Program

Example: nano ParentChild.c

Step 10: Compile the Program

```
saisharan134@APL-5CD1366K5L:~$ gcc ParentChild.c -o ParentChild
saisharan134@APL-5CD1366K5L:~$ ./ParentChild
PID: 895 Count: 1
PID: 896 Count: 1
PID: 895 Count: 2
PID: 896 Count: 2
PID: 895 Count: 3
PID: 896 Count: 3
PID: 895 Count: 4
PID: 896 Count: 4
PID: 895 Count: 5
PID: 896 Count: 5
```

GitHub Repository Link:



Fork Programmes:

Create Fork.c

```
GNU nano 8.7.1 Fork.c
#include<stdio.h>
#include<unistd.h>

int main()
{
    fork();

    printf("Hello\n");

    return 0;
}
```

Output:

```
saisharan134@APL-5CD1366K5L:~$ nano Fork.c
saisharan134@APL-5CD1366K5L:~$ gcc Fork.c -o Fork
saisharan134@APL-5CD1366K5L:~$ ./Fork
Hello
Hello
saisharan134@APL-5CD1366K5L:~$ |
```

Create ParentAndChild.c

```
GNU nano 8.7.1 ParentAndChild.c *
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else
    {
        printf("I am Parent Process\n");
    }
    return 0;
}
```

Output:

```
saisharan134@APL-5CD1366K5L:~$ nano ParentAndChild.c
saisharan134@APL-5CD1366K5L:~$ gcc ParentAndChild.c -o ParentAndChild
saisharan134@APL-5CD1366K5L:~$ ./ParentAndChild
I am Parent Process
I am Child Process
```

